

María Paula Silva Arévalo

Ph.D. Student in Physics

University of Delaware, Newark, DE, USA

BSc. Physics

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ABOUT ME

I am a Ph.D. student in Physics at the University of Delaware, with research interests in stellar variability, exoplanet detection, planetary and stellar atmospheres. My current work focuses on developing and applying statistical tools to characterize stellar activity in time series data, including radial velocities, activity indicators, and photometry. I am particularly interested in how stellar variability influences the detection and interpretation of exoplanet signals, and how understanding host star behavior can improve studies of planetary atmospheres, habitability, and potential biosignatures.

EDUCATION

Ph.D. in Physics

University of Delaware, Newark, DE, USA

Started: August 2025

Bachelor of Science in Physics - Universidad Nacional de Colombia, Bogotá

Graduation Date: 12/12/2024

CGPA: 4.2/ 5.0

LANGUAGES

English – C1

German – B1

PROGRAMMING LANGUAGES

PYTHON (*Upper-Intermediate*)

-Libraries: Numpy, Matplotlib, Scipy, Astropy, Pandas

Research Experience

Graduate Research: Persistence Time in Stellar Atmospheres

University of Delaware, Newark, DE, USA

August 2025 – Present

Supervised by Prof. Sarah Dodson-Robinson

Developed and tested an adapted implementation of the TAUEST algorithm to estimate persistence times in stellar atmosphere observables. This project applies autoregressive modeling and bias-correction techniques to stellar time series, including radial velocities, activity indicators, and photometry, with the goal of characterizing stellar variability and improving stellar-noise modeling for exoplanet detection. Released the adapted TAUEST implementation as a publicly available Python package hosted in a GitHub repository, allowing researchers to apply the method to their own stellar time series datasets.

Graduate Research: Extreme Stellar Signals IV

University of Delaware, Newark, DE, USA

August 2025 – November 2025

Part of the University of Delaware team for the ESSP IV. Project on high-resolution spectrograph data of the sun as a star to test different techniques for separating stellar signals from synthetic planetary signals injected to the data.

Research Internship: Implementation of Statistical Methods for the Study of Stellar Atmosphere Signals.

University of Delaware, Newark, Delaware, US

June 2024 – November 2024

Supervised by prof. Sarah Dodson-Robinson

Exploration of a new planetary model for 47 UMa using radial velocity data. This model includes two planets orbiting the star and shows that the third confirmed planet does not meet the Rayleigh resolution criterion.

Undergraduate Thesis Research: Seasonal Variation in Martian Irradiance and its Influence on the Physiological Behavior and Growth of Chenopodium quinoa

National University of Colombia, Bogotá, Colombia

October 2023 – June 2024

Supervised by prof. Santiago Vargas Dominguez

- Investigated and compared the seasonal variation in the distribution of radiant energy on Mars' surface.
- Designed controlled experiments to evaluate the impact of varying levels of irradiance on the physiological behavior and growth of Chenopodium quinoa and analyzed how light intensity affected fundamental biological processes in Chenopodium quinoa, such as photosynthesis and growth rates, under simulated Martian conditions.

PUBLICATIONS & PRESENTATIONS

Publication Efficiency in Astronomy: A Global GDP Based Comparison (2019–2023)

Research Notes of the AAS, 2025

DOI: 10.3847/2515-5172/adf74c

Authors: S. Guevara-Montoya, M. P. Silva-Arévalo, F. Ortiz-Ferreira, J. E. Forero-Romero

Astronomy in Colombia: A Bibliometric Perspective

Journal of the Colombian Academy of Exact, Physical, and Natural Sciences, 2024.

DOI: 10.18257/raccefyn.2589

Authors: S. Guevara-Montoya, F. Ortiz-Ferreira, M. P. Silva-Arévalo, P. A. Niño-Muñoz, J. E. Forero-Romero

Measuring Persistence Time in Stellar Atmospheres (In preparation)

Authors: M. P. Silva-Arévalo, S. Dodson-Robinson, A. Ejaz, J. Harrell

Poster presented at the APS Mid-Atlantic Section Meeting, Philadelphia, PA, USA, November, 2024.

Poster presented at Exoplanets 6, Porto, Portugal, June 29–July 3, 2026.

Poster presented at HORS3S, Granada, Spain, July 6–10, 2026.